

TABLE 2
THE STATUS OF MAJOR SEMICONDUCTOR MANUFACTURING PROJECTS IN THE US

Company	Location	Investment Amount	Project Details	Expected Completion Date
Intel Corporation	Chandler, Arizona	Not specified	Construction of two new fabs, Fab 52 and Fab 62.	2024
	Licking County, Ohio	\$20 billion	Construction of two fabs. Recent reports indicate that the production facilities are now slated for completion in 2030 and 2031, extending beyond the initially promised 2028 deadline.	2030 and 2031
Taiwan Semiconductor Manufacturing Company	Phoenix, Arizona	Additional \$100 billion	The expansion includes the construction of three additional fabs, two advanced packaging plants, and a research and development centre. The first Arizona factory has begun mass production of 4-nanometer chips.	Not specified
Samsung Electronics	Taylor and Austin, Texas	Approximately \$40 billion	Construction of two new semiconductor fabrication plants and a research and development facility in Taylor, as well as an expansion of the existing facility in Austin.	Not specified
Micron Technology	Clay, New York, and Boise, Idaho	Not specified	Construction of new semiconductor manufacturing facilities, including a “mega-fab” in Clay, New York, expected to create approximately 20,000 jobs.	Not specified
GlobalFoundries	Malta, New York, and Essex Junction, Vermont	\$1.5 billion	Building a new fab in Malta and upgrading the Essex Junction plant to enhance the production of critical semiconductors for various sectors.	Not specified
Integra Technologies	Wichita, Kansas	\$1.8 billion	Proposal to build a one-million-square-foot OSAT facility and headquarters, expected to create at least 2000 new jobs.	Not specified
Amkor Technology	Peoria, Arizona	\$2 billion	Plans to build an advanced packaging and testing facility, motivated by collaborations with companies like Apple and TSMC.	Not specified

(15.24cm) SiC power semiconductors at its facility in Japan.

On January 14, 2025, Infineon began constructing its new semiconductor backend production facility in Samut Prakan, south of Bangkok, Thailand. The first phase of the facility is expected to start operations by early next year, with plans for flexible capacity expansions based on market demand.

Indichip Semiconductors Ltd, an emerging power semiconductor startup in Amaravati, India, has partnered with Japan’s Yitoo Micro Technology (YMTL) to invest ₹140 billion (around \$1.6 billion) in establishing a silicon carbide wafer manufacturing plant in Andhra Pradesh.

The initial production capacity is expected to reach 10,000 wafers per month, with plans to scale up to 50,000 wafers per month within two to three years. Initially focusing on 6-inch (15.24cm) SiC wafers, the company plans to transition to 8-inch (20.32cm) wafers.

Furthermore, Fuji Electric has started mass production at its

facility in Aomori Prefecture, Japan, as of December 2024. The company had initially planned to begin production in the summer of 2024, but delays due to lower global demand for EVs pushed back the schedule.

Besides, China’s influence in the semiconductor industry is undeniable, with major players deepening their regional investments. At GlobalFoundries’ 2024 summit in Shanghai, the company clarified that its future strategy hinges on local partnerships rather than setting up its own facilities.

Likewise, STMicroelectronics has strengthened its presence through key collaborations with Sanan Optoelectronics and Hua Hong Semiconductor. European firms like Infineon and NXP are also ramping up their ‘Made in China’ efforts, recognising that despite geopolitical tensions, China’s demand remains crucial for growth.

Why the West is at the heart of these delays

The semiconductor story is not just about technology—it is about

power, politics, and shifting global dynamics. Production spans several specialised stages—design, fabrication, assembly, testing, and packaging—each distributed across different regions.

US and European firms dominate semiconductor design, with Nvidia leading in AI chips and Arm’s technology embedded in most consumer electronics. Taiwan’s TSMC manufactures 60% of the world’s chips and 90% of advanced semiconductors, while South Korea remains a key supplier of memory components. Assembly, testing, and packaging are concentrated in China, Japan, and South Korea, reinforcing East Asia’s dominance.

This regional concentration makes the supply chain highly vulnerable. A disruption in one region can ripple across industries worldwide, intensifying the urgency for diversification as geopolitical risks escalate.

With Joe Biden’s term having ended on January 20, 2025, uncertainty looms over the US CHIPS Act of 2022 and the return of Trump’s